

Deep Learning

Week 13 – Part 1 – Deep Learning Basics

CS 457 - L1 Data Science

Zeesham Rasheed

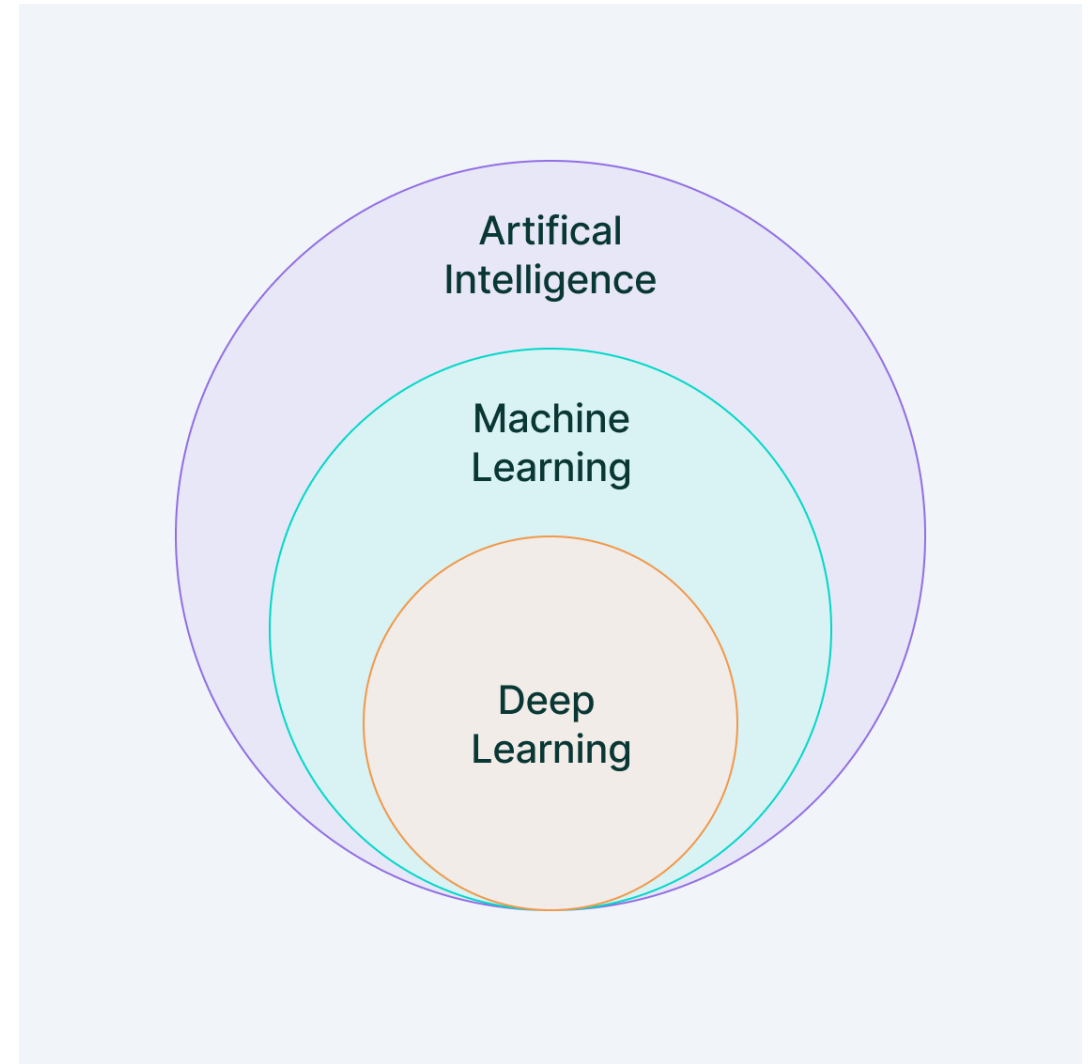
- **Artificial Intelligence (AI)**
- **Machine Learning (ML)**
- **Deep Learning (DL)**

- **AI** is intelligence demonstrated by machines, as opposed to natural intelligence displayed by animals including humans.
- **ML** is the study of computer algorithms that can improve automatically through experience and using data. It is seen as a part of artificial intelligence.
- **DL** is part of a broader family of machine learning methods based on Artificial Neural Networks (ANN)

Relationship of AI, ML and DL



- Artificial Intelligence (AI) is anything about man-made intelligence exhibited by machines.
- Machine Learning (ML) is an approach to achieve AI.
- Deep Learning (DL) is **one technique** to implement ML

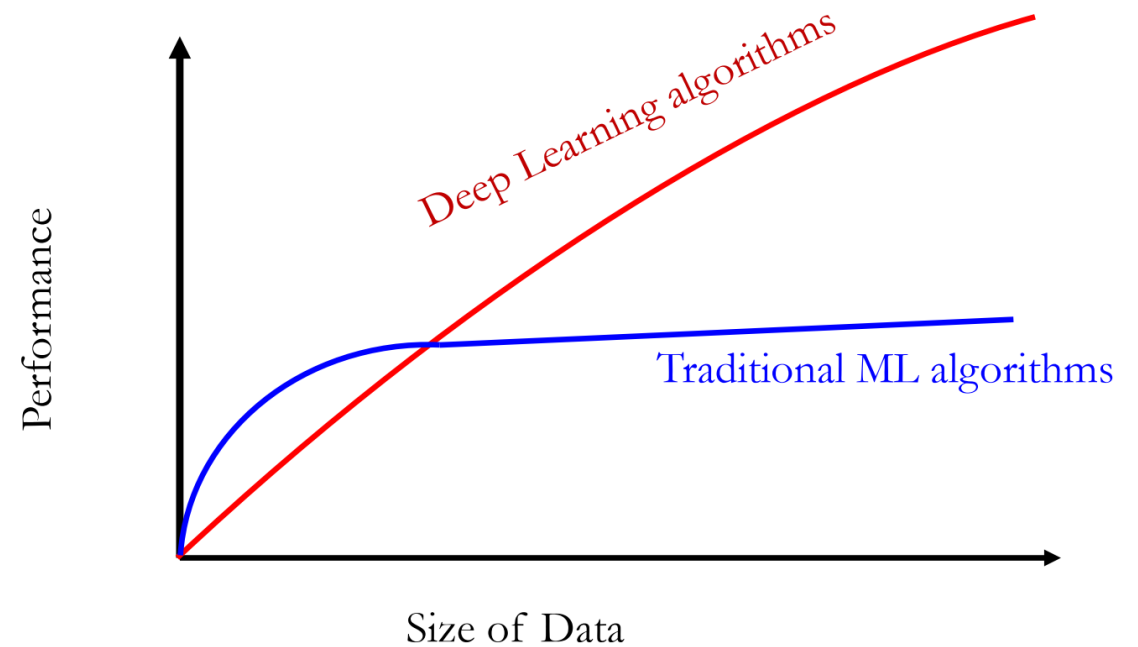


- Deep Learning is a part of machine learning that deals with algorithms inspired by the structure and function of the human brain.
- It uses artificial neural networks to build intelligent models and solve complex problems.
 - Deep learning is a recent buzz word for neural networks.
- Deep learning is mostly used with unstructured data like text and images

Why Deep Learning?



- Limitations of traditional machine learning algorithms
 - not good at handling high dimensional data.
 - difficult to do feature extraction and object recognition.
- Advantages of deep learning
 - DL is computationally expensive, but it is capable of handling high dimensional data.
 - **Feature extraction is done automatically (image and text data)**



Deep Learning vs Machine Learning



Machine Learning	Deep Learning
Apply statistical algorithms to learn the hidden patterns and relationships in the dataset.	Uses artificial neural network architecture to learn the hidden patterns and relationships in the dataset.
Can work on the smaller amount of dataset	Requires the larger volume of dataset compared to machine learning
Better for the low-label task.	Better for complex task like image processing, natural language processing , etc.
Takes less time to train the model.	Takes more time to train the model.
A model is created by relevant features which are manually extracted from images to detect an object in the image.	Relevant features are automatically extracted from images. It is an end-to-end learning process.
Less complex and easy to interpret the result.	More complex, it works like the black box interpretations of the result are not easy.
It can work on the CPU or requires less computing power as compared to deep learning.	It requires a high-performance computer with GPU.

Feature Extraction

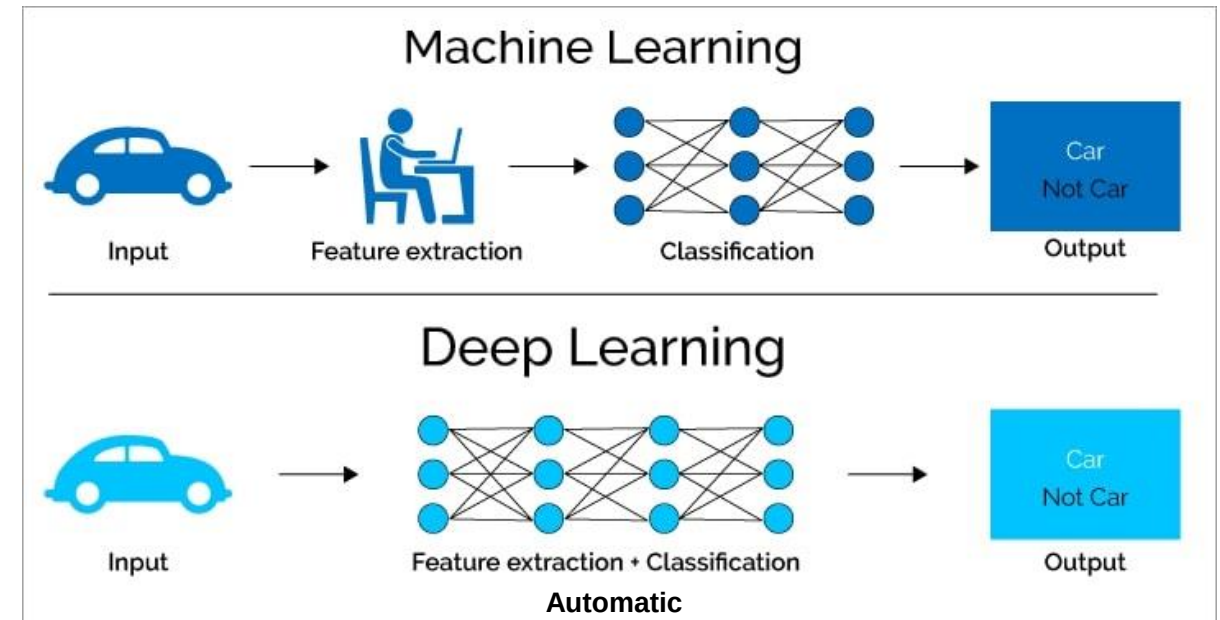


- **Machine Learning**

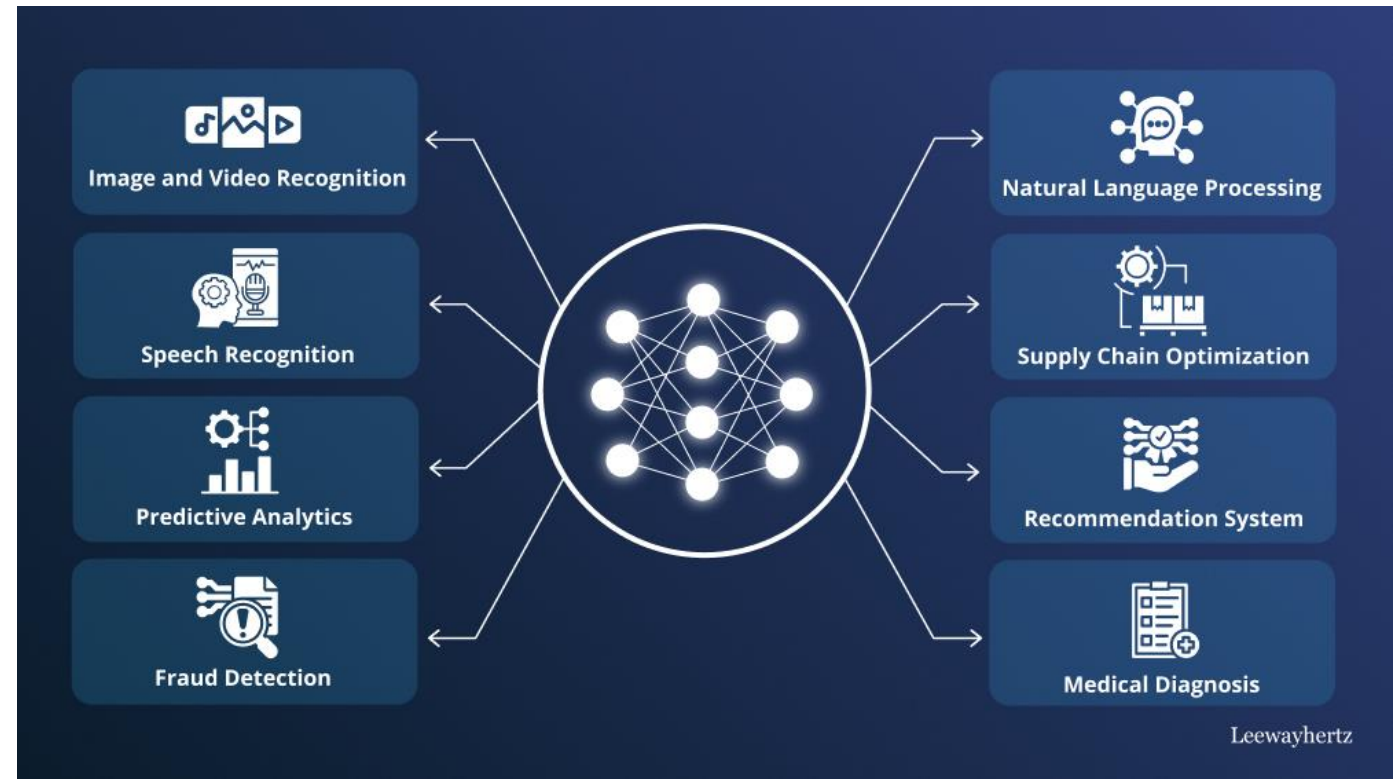
- Need to provide inputs like shape, color, size etc
- Imagine input features (columns) in your datasets

- **Deep Learning**

- Input a text document or image
- DL will extract features automatically
- DL learns representation of data itself



- The main applications of deep learning AI can be divided into **computer vision** (images), **natural language processing** (text), and **Reinforcement learning**.
- Reinforcement learning
 - an interesting domain of artificial intelligence that simulates the learning process by trial and error
 - mimicking how humans and animals learn from the consequences of their actions
 - used in Recommendation Systems and Game Development



- In computer vision, Deep learning AI models can enable machines to identify and understand visual data.
- Some of the main applications of deep learning in computer vision include
 - **Object detection and recognition:** Deep learning model can be used to identify and locate objects within images and videos, making it possible for machines to perform tasks such as self-driving cars, surveillance, and robotics.
 - **Image classification:** Deep learning models can be used to classify images into categories such as animals, plants, and buildings. This is used in applications such as medical imaging, quality control, and image retrieval.
 - **Image segmentation:** Deep learning models can be used for image segmentation into different regions, making it possible to identify specific features within images.

- In NLP, the Deep learning model can enable machines to understand and generate human language.
- Some of the main applications of deep learning in NLP include:
 - **Automatic Text Generation:** Deep learning model can learn the corpus of text and new text like summaries, essays can be automatically generated using these trained models.
 - **Language Translation:** Deep learning models can translate text from one language to another, making it possible to communicate with people from different linguistic backgrounds.
 - **Sentiment analysis:** Deep learning models can analyze the sentiment of a piece of text, making it possible to determine whether the text is positive, negative, or neutral. This is used in applications such as customer service, social media monitoring, and political analysis.
 - **Speech recognition:** Deep learning models can recognize and transcribe spoken words, making it possible to perform tasks such as speech-to-text conversion, voice search, and voice-controlled devices.

Advantages of Deep Learning



- **High accuracy:** Deep Learning algorithms can achieve state-of-the-art performance in various tasks, such as image recognition and natural language processing.
- **Automated Feature Engineering:** Deep Learning algorithms can automatically discover and learn relevant features from data without the need for manual feature engineering.
- **Scalability:** Deep Learning models can scale to handle large and complex datasets and can learn from massive amounts of data.

Disadvantages of Deep Learning



- **High computational requirements:** Deep Learning AI models require large amounts of data and computational resources to train and optimize.
- **Requires large amounts of labeled data:** Deep Learning models often require a large amount of labeled data for training, which can be expensive and time-consuming to acquire.
- **Interpretability:** Deep Learning models can be challenging to interpret, making it difficult to understand how they make decisions.

End of Part 1



Deep Learning

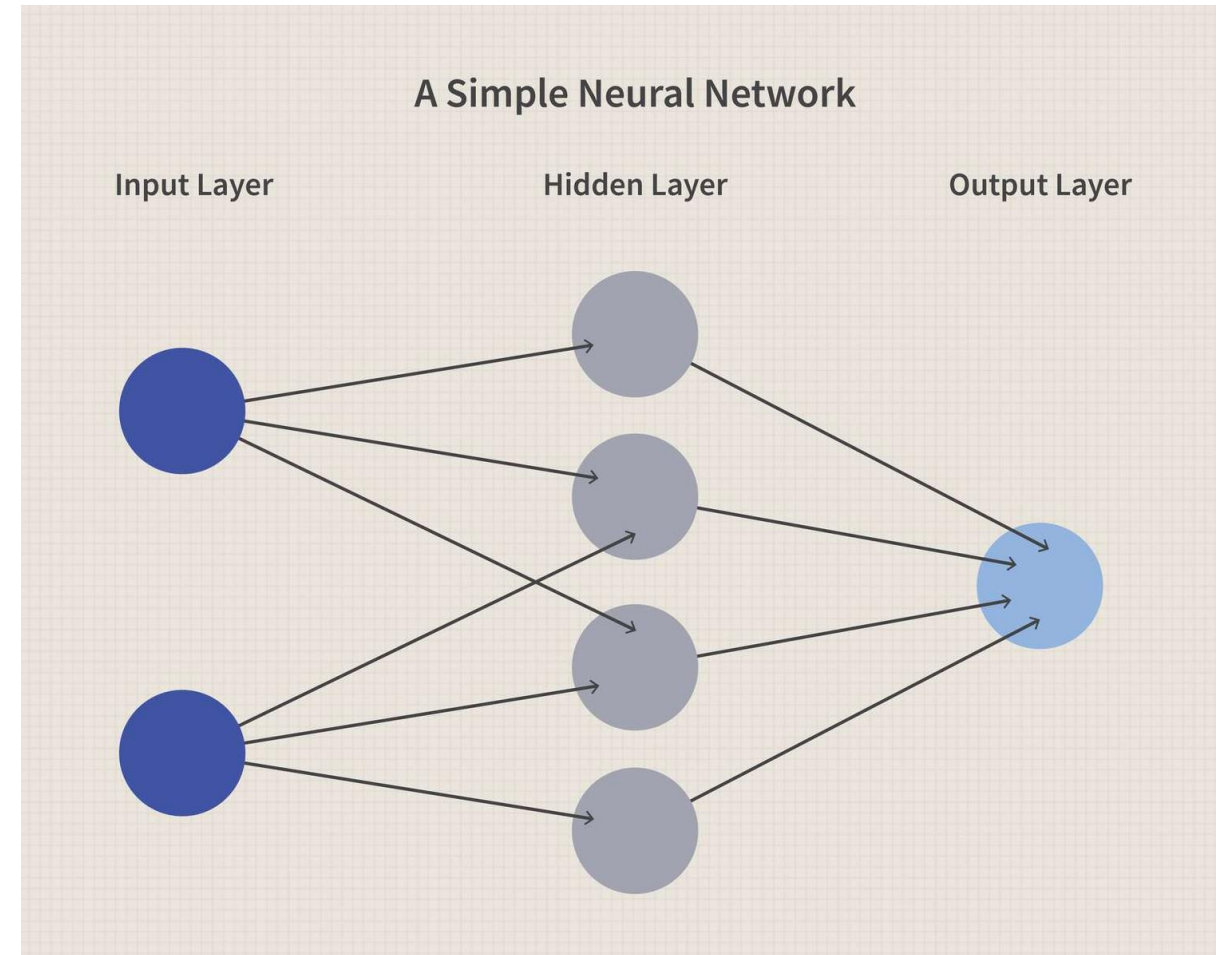
Week 13 – Part 2 – Neural Networks

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- Neural networks, also called artificial neural networks (ANNs) are a subset of machine learning and are the backbone of deep learning algorithms.
- They are called “neural” because they mimic how neurons in the brain signal one another.
 - People get a lot of ideas from nature
 - For example, airplanes were invented inspired by birds.
 - A neural network is a type of machine-learning algorithm that is inspired by the human brain. It is a network of interconnected nodes, or artificial neurons, that learn to recognize patterns in data.

- A neural network has three main layers.
 - **Input Layer:** This layer is responsible for accepting the inputs.
 - **Hidden Layer:** This layer processes the input data to find out hidden information and performs feature extraction.
 - There can be multiple hidden layers
 - **Output Layer:** This layer gives the desired output.



- The input layer acts as the conduit for input data, translating raw information into a format the network can process. The layer comprises multiple nodes, each node representing one feature of the input data. This layer is the entry point for data into the neural network.
- It takes raw data and passes it to the next layer for processing.

- Hidden layers perform the heavy lifting of feature extraction and pattern recognition. Each layer builds upon the previous one, capturing increasingly complex data representations.
- These layers are termed “hidden” because they do not have direct interactions with the input or output; they are situated between the input and output layers.
- Multiple hidden layers process information at multiple levels of abstraction. Each layer extracts and transforms features from the previous layer, enabling the network to understand complex patterns and relationships in the data.

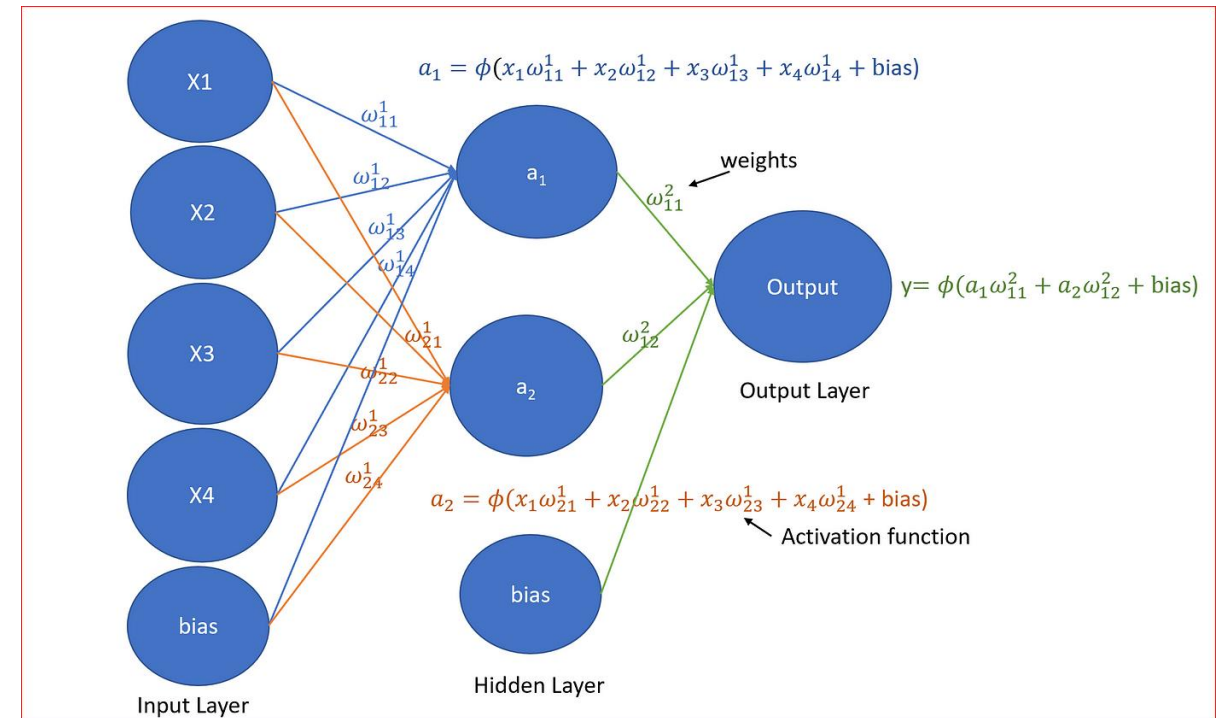
- The final stage of the network is where processed information is converted into a human-understandable format, such as a classification label or a numerical value. The output layer comprises nodes that provide the final results of the neural network's processing.
- It delivers the output of the network, which can vary depending on the task:
 - Binary classification: The output layer might have two nodes for tasks with a “yes” or “no” answer.
 - Multiclass classification: The output layer will have more nodes for tasks with multiple possible outcomes, each representing a different class or category.

Neural Network Calculations



The following operations are performed within each neuron

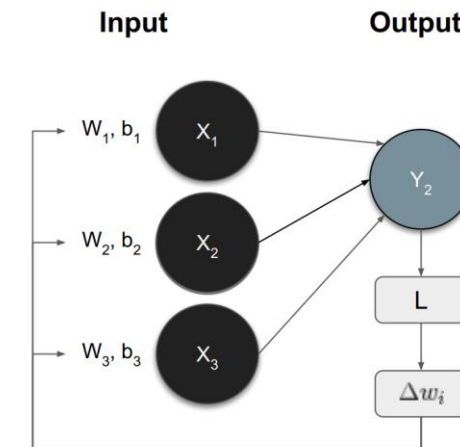
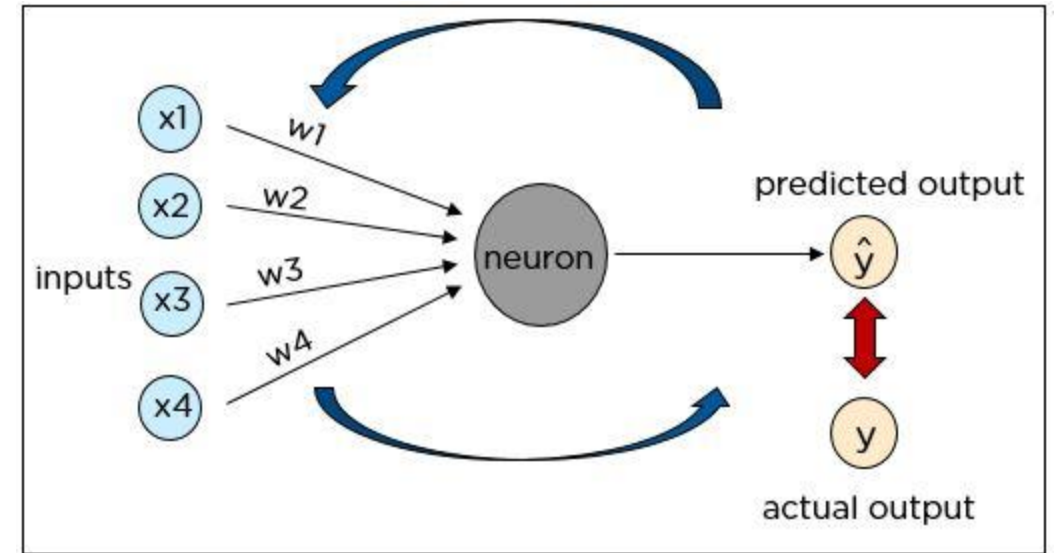
- The product of each input value and the weight of the channel (initially randomly assigned) has passed over is found.
- It computes the sum of the weighted products. We call this the **weighted sum**.
- Then it adds a **bias** unique to the neuron to the weighted sum.
 - Bias in a neural network is a constant value that's added to the product of inputs and weights to adjust the output of a neuron
- We then subject the final sum to a particular function called **activation function**
 - There are number of activation functions available in a neural network. For example, **Sigmoid Function**, which converts the calculation into probability of outcomes (ranges from 0-1)
- This process keeps iterating, every time adjusting the weights and bias in order to minimize error and train best model.



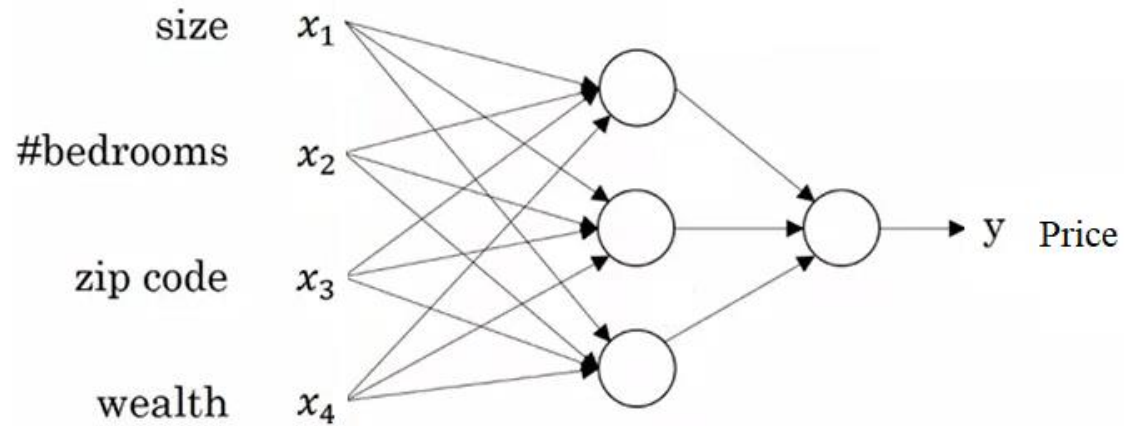
Neural Network – ML Model Training



- The **Cost function** returns the difference between the neural network's predicted output and the actual output from a set of labeled training data.
- The goal is to **minimize the cost function** that can be obtained by adjusting the weights and biases iteratively throughout the training process.
- The formula for cost function is:
 - $Cost = Y - \hat{Y}$
 - Where $\hat{Y}$ is the predicted value and Y is the actual output.

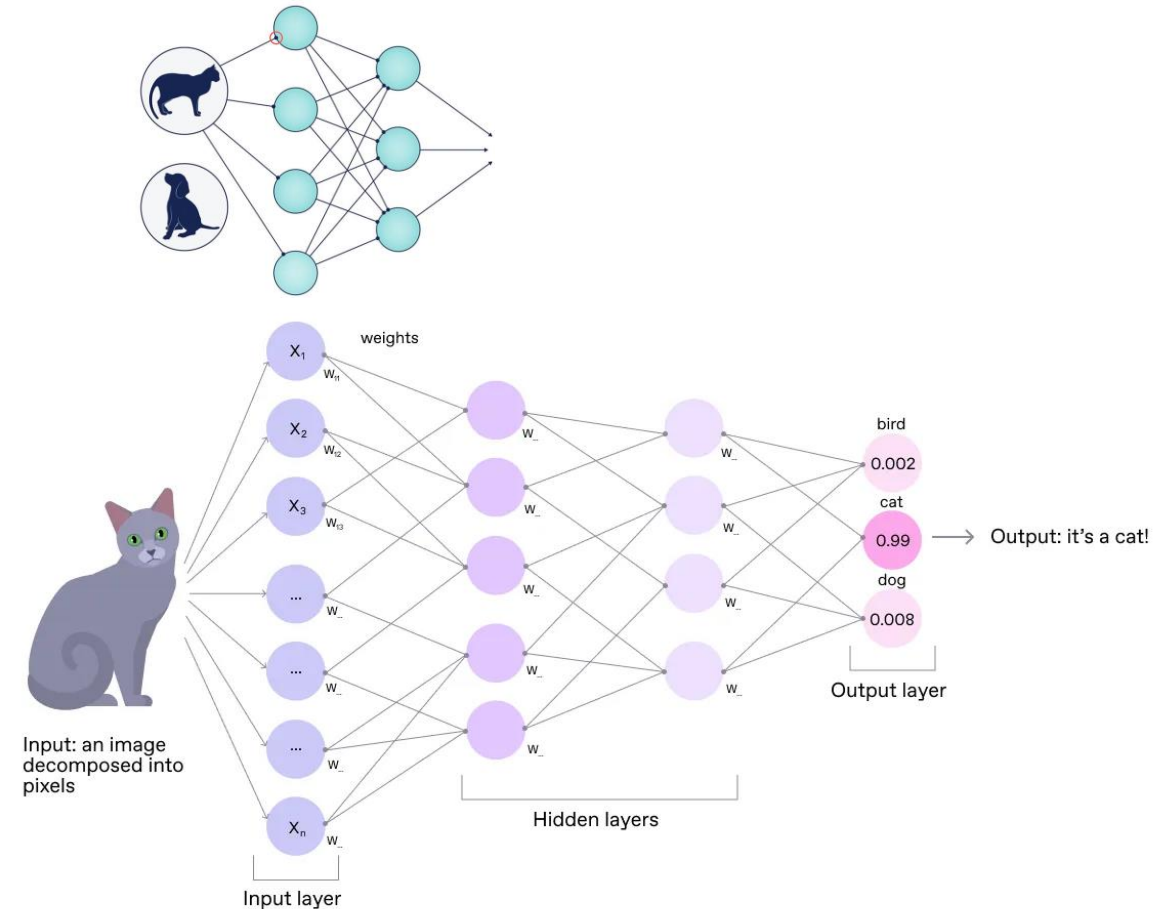


- Housing Price Prediction



- Image Classification

- Training NN to classify Cat or Dog



When Neural Network becomes Deep Learning

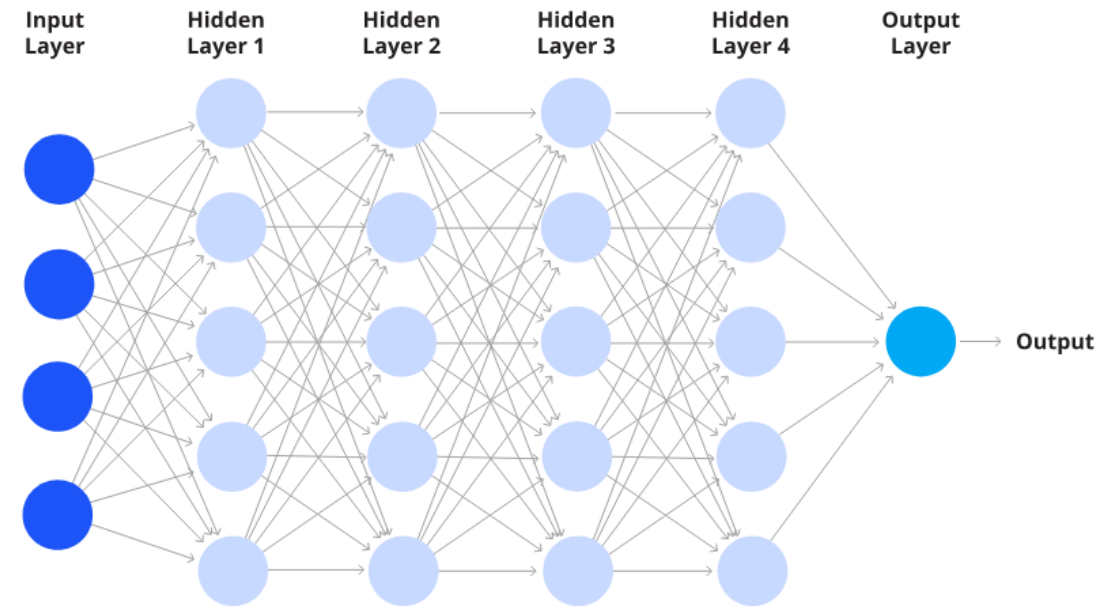


- The term, Deep Learning, refers to training Neural Networks, **sometimes very large Neural Networks**.
- The “**deep**” in deep learning refers to the **depth of layers (hidden Layers)** in a neural network.
 - A neural network of **more than three layers**, including the inputs and the output, can be considered a deep-learning algorithm.
- While conventional Machine Learning algorithms are limited to relatively superficial analysis, deep learning models can extract richer, more abstract representations from large and complex datasets like images, text, and audio.
- **Applications:** Deep learning underpins today’s rapid advances in Artificial Intelligence, particularly in computer vision, speech recognition and natural language processing

Typical Deep Learning Architecture



- Stacking large numbers of hidden layers between the input and the output allows your neural network to perform deep-learning tasks on the input
- A deep learning network's power comes from its ability to **learn and refine features through its hidden layers**, ultimately producing accurate and meaningful outputs based on the input data



End of Part 2



Deep Learning

Week 13 – Part 3 – Image Data

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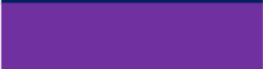
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Image Representation



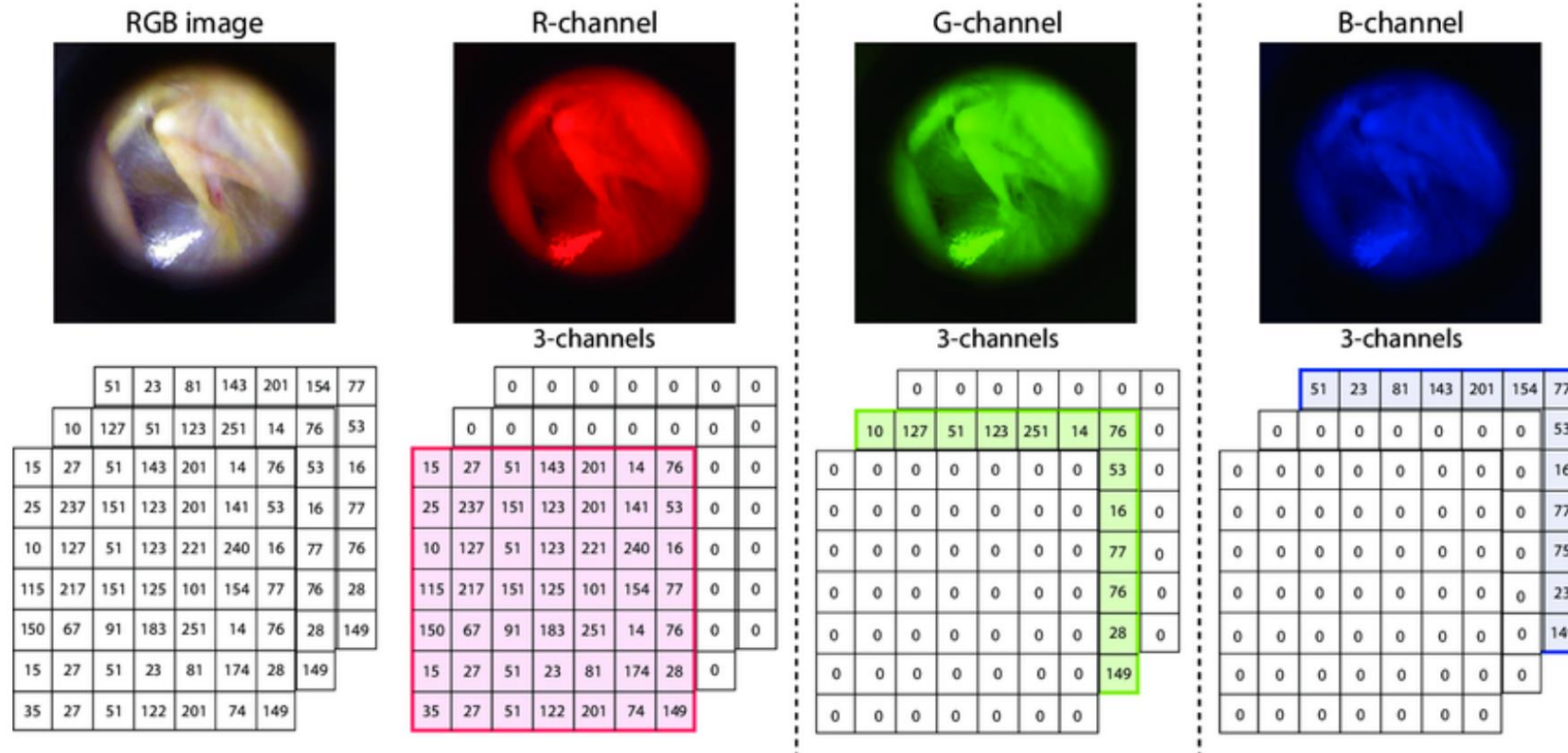
- **Pixels and Colors**

- An image is made of many pixels, each of them with a particular color.
- Modern computers can show up to 16.7 millions of different colors.
- Each color is represented as a combination of three primary colors: Red, Green and Blue (**RGB color model**).
- Therefore, for each pixel, we need to store a total of 3 numbers corresponding to the amount of each color.
 - For most computers nowadays, each color can get 256 values (from 0 to 255), 0 being the absence of the color and 255 the highest intensity of it.

Color		R	G	B
Dark Red		192	0	0
Red		255	0	0
Orange		255	192	0
Yellow		255	255	0
Light Green		146	208	80
Green		0	176	80
Light Blue		0	176	240
Blue		0	112	192
Dark Blue		0	32	96
Purple		112	48	160
White		255	255	255
Black		0	0	0

Color Image

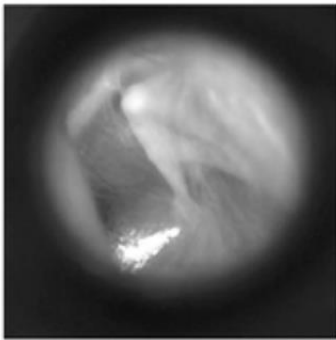
- Every color image (RGB image) has three channels (red, green, blue) when you read and process it in your computer.
- Pixel value ranges from 0 to 255



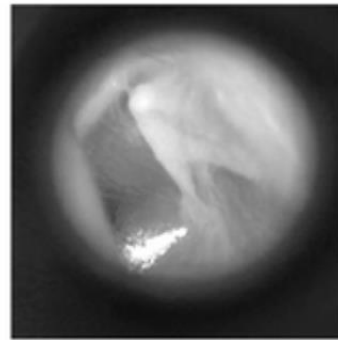
Grayscale Image

- Grayscale image has 1 channel when you read and process it in your computer.
- Pixel value ranges from 0 to 255

Grayscale image

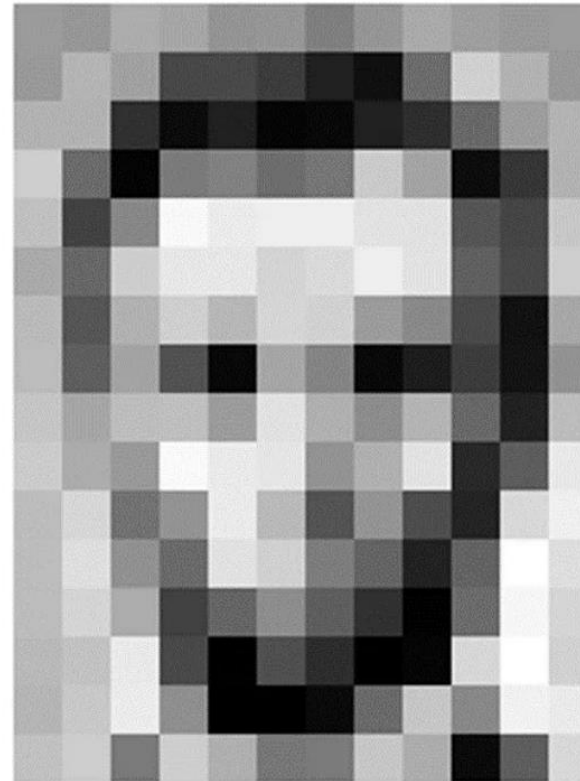


15	27	51	143	201	14	76
25	237	151	123	201	141	53
10	127	51	123	221	240	16
115	217	151	125	101	154	77
150	67	91	183	251	14	76
15	27	51	23	81	174	28
35	27	51	122	201	74	149



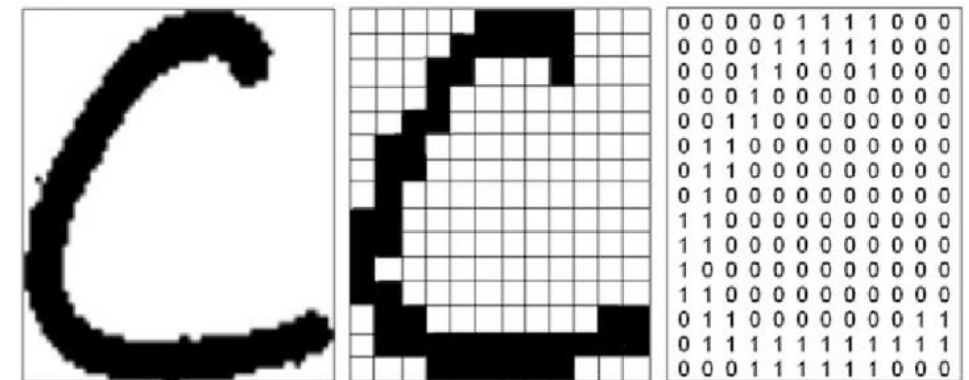
1-channel

15	27	51	143	201	14	76
25	237	151	123	201	141	53
10	127	51	123	221	240	16
115	217	151	125	101	154	77
150	67	91	183	251	14	76
15	27	51	23	81	174	28
35	27	51	122	201	74	149



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	105	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	105	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	292	237	145	0	0	12	108	209	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

- [illegible]



End of Part 3



Deep Learning

Week 13 – Part 4 – Image Classification
using Deep Learning

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The Success of Neural Networks



- Among deep neural networks (DNN), the convolutional neural network (CNN) has demonstrated excellent results in computer vision tasks, especially in image classification.
- Convolutional Neural Networks (CNNs) are a special type of multi-layer neural network inspired by the mechanism of human optical and neural systems.

Convolutional Neural Network (CNN)



- A CNN is a framework developed using machine learning concepts. CNNs can learn and train from data on their own without the need for human intervention.
- There is only some pre-processing needed when using CNNs.
 - They develop and adapt their image filters (or kernels), which have to be carefully coded for most algorithms and models.
 - These filters/kernels help CNN creating features automatically
 - CNN frameworks have a set of layers that perform particular functions to enable CNN to perform these functions.

Difference between NN and CNN

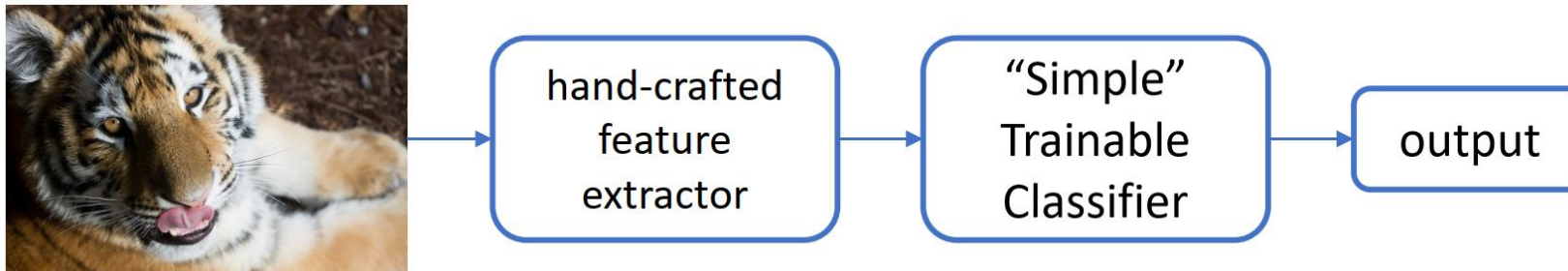


- The main differences between neural networks (NNs) and convolutional neural networks (CNNs) are
 - the type of data they work with
 - the layers in their architecture
 - and their intended use cases
- Data type
 - NNs can work with tabular or text data, while CNNs work with both text and image data.
- Layers
 - CNNs have three main layers: **convolutional, pooling, and fully-connected layers**

Traditional ML Models for Images



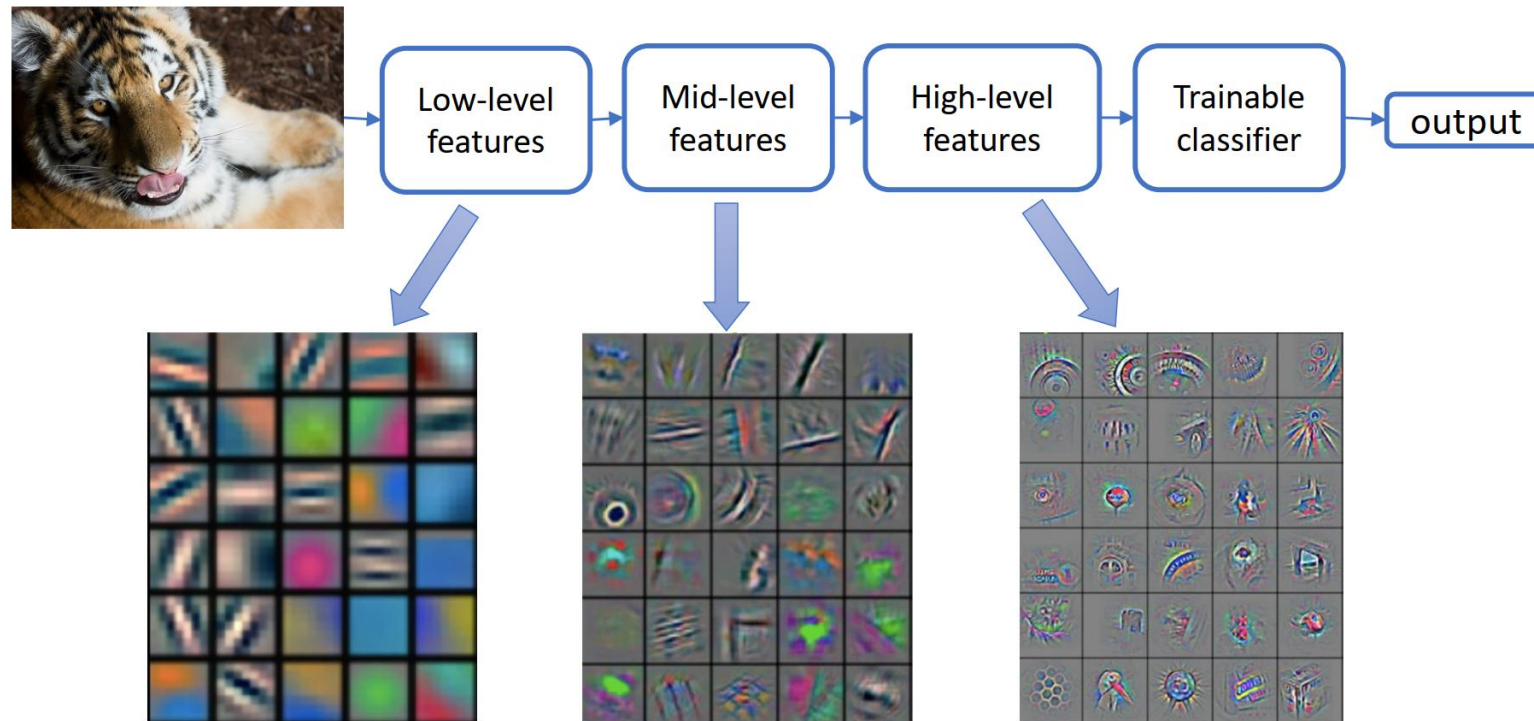
- Traditional pattern recognition models use hand-crafted features and relatively simple trainable classifier.
- This approach has the following limitations:
 - It is very tedious and costly to develop hand-crafted features
 - The hand-crafted features are usually highly dependents on one application, and cannot be transferred easily to other applications



Deep Learning Models for Images



- Deep learning (representation learning) seeks to learn rich hierarchical representations (features) automatically through multiple stage of feature learning process

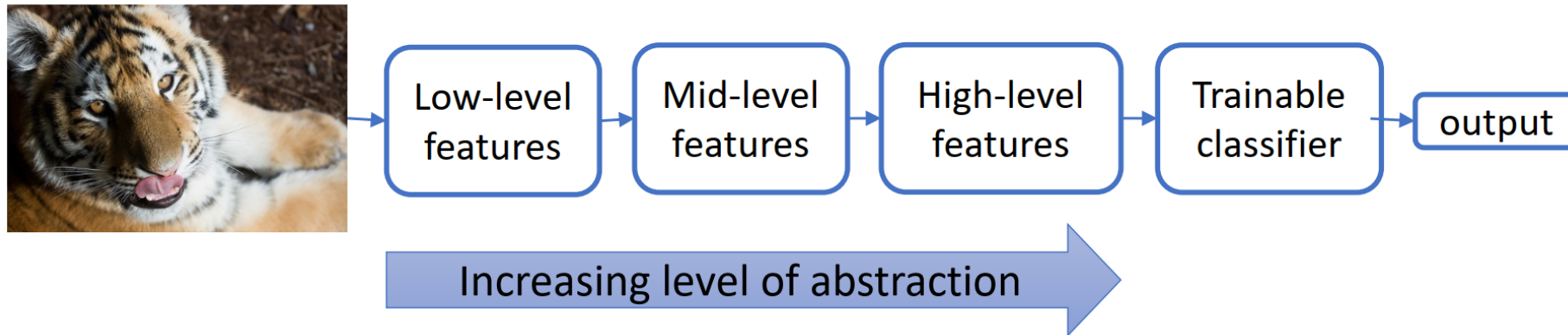


Feature visualization of convolutional net trained on ImageNet
(Zeiler and Fergus, 2013)

Hierarchical Representation

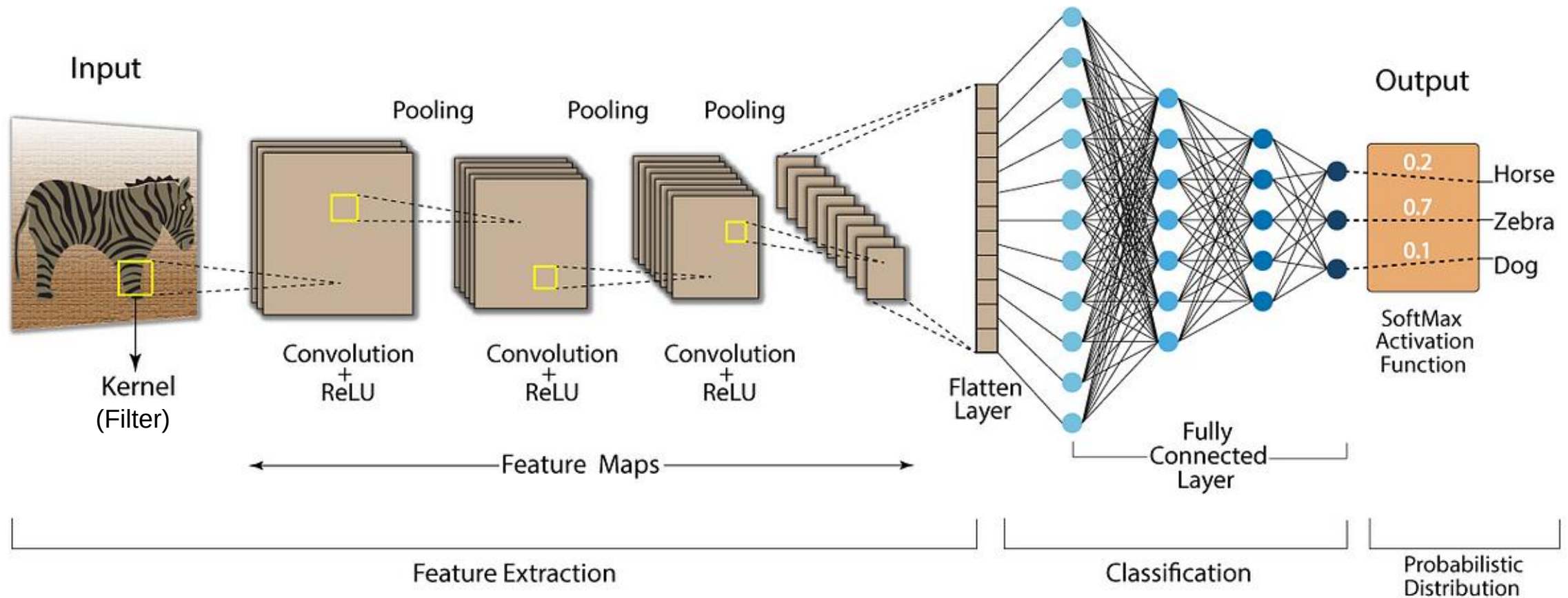


- Hierarchy of representations with increasing level of abstraction. Each stage is a kind new feature transformation
- **Image recognition**
 - Pixel → edge → texton → motif → part → object
- **Text**
 - Character → word → word group → clause → sentence → story



- A CNN system may have somewhere between 3 to 150 or even more layers
 - The “deep” of Deep neural networks refers to the number of layers.
 - One layer’s output acts as another layer’s input.
 - Deep multi-layer neural networks include Resnet50 (50 layers) or ResNet101 (101 layers).

CNN Architecture for Image Classification



- **Convolution Layer:** A convolution is the simple application of a filter to an input (image). The convolution layer has a set of trainable filters. Convolution layers are the major building blocks used in convolutional neural networks finding patterns and creating features
 - ReLu Layer: ReLu layers, or Rectified linear unit layers, are activation functions for enhancing features in training and building CNN.
- **Pooling Layer:** This layer collects the result of all neurons in the layer preceding it and processes this data. The primary task of a pooling layer is to lower the number of considered features and give streamlined output.
- **Fully-Connected Layer:** This layer is the final output layer for CNN models that flattens the input data received from layers before it and gives the result.

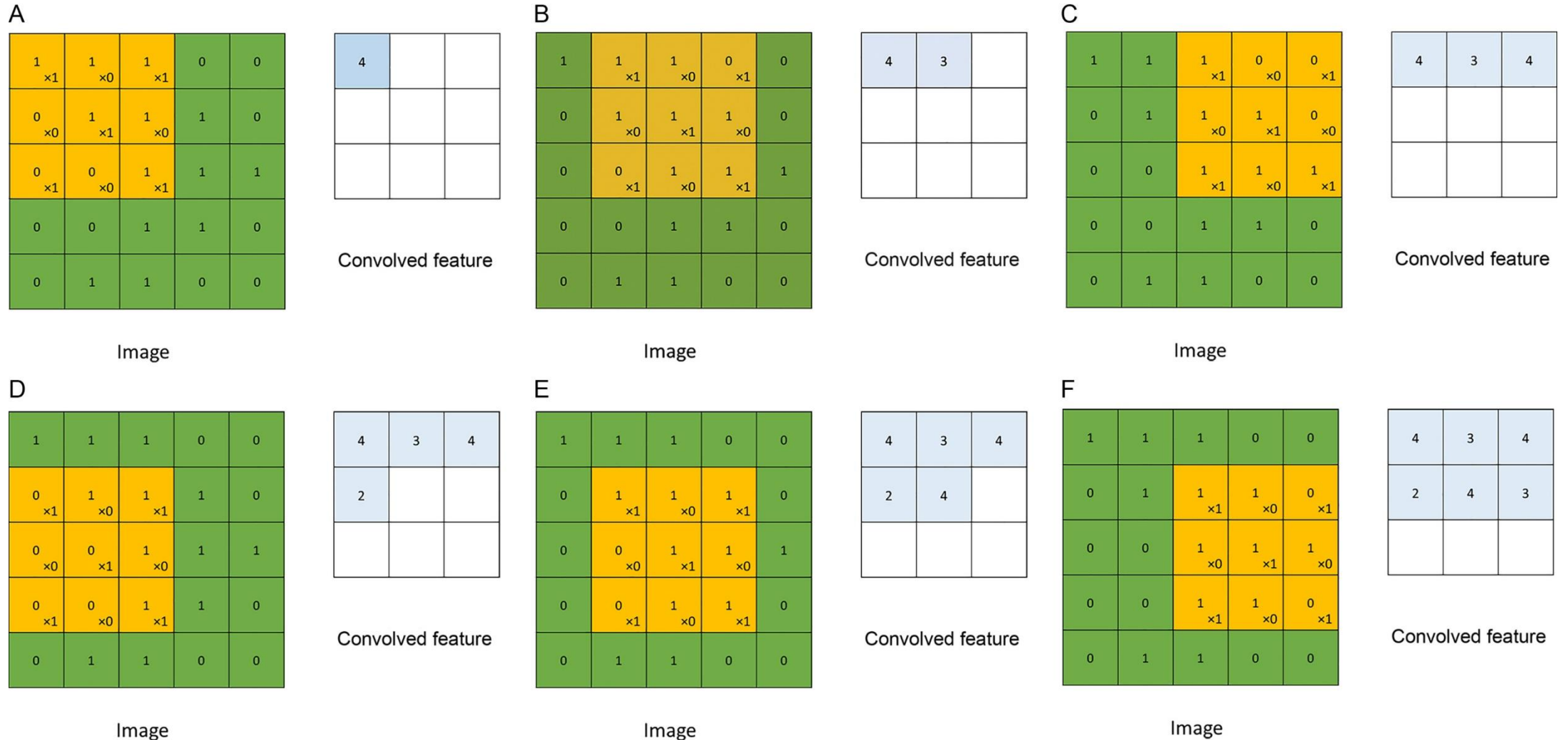
Convolution Layer Operation



- Convolution operation is element wise multiplication and addition.
- Convolution operation is used to produce/create **input features**



Convolution Layer - Example



Different Filters/Kernels



- Different filters/kernels producing Feature Maps
- We learn patterns from these filters.



Original



Sharpen



Edge Detect



“Strong” Edge
Detect

Another Convolution Filter Example



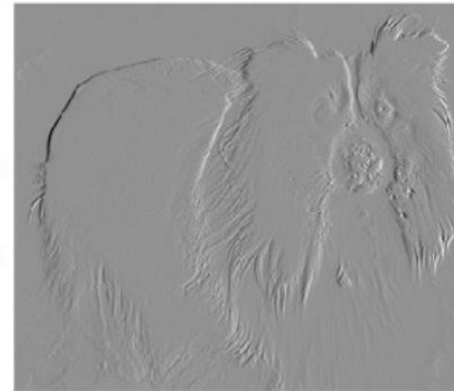
- One simple filter/kernel to emboss and detect edges



Input



Filter



Output

Feature Map

CNN – Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

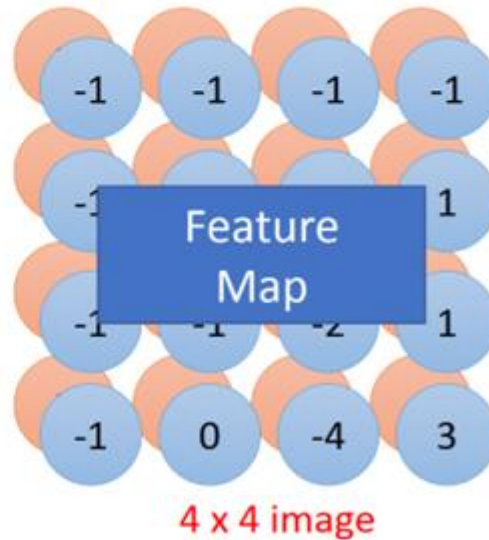
Filter 2

stride=1

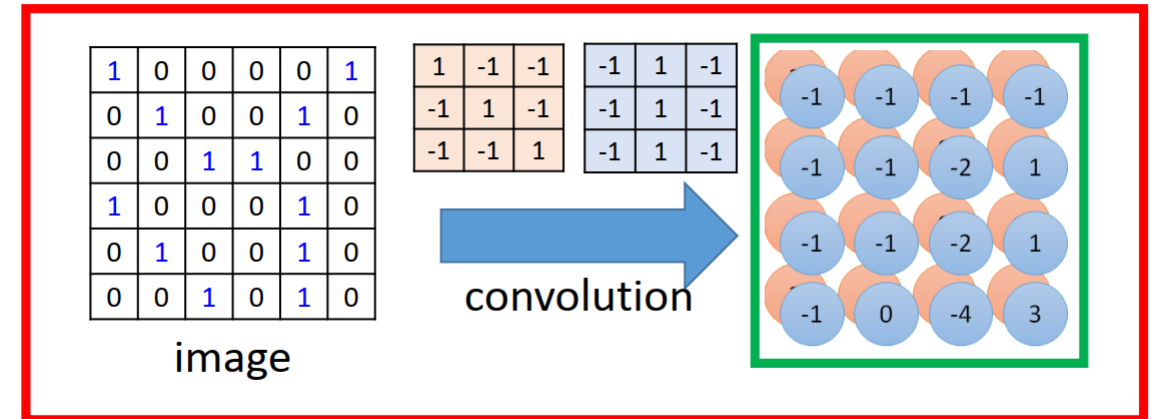
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Do the same process for every filter



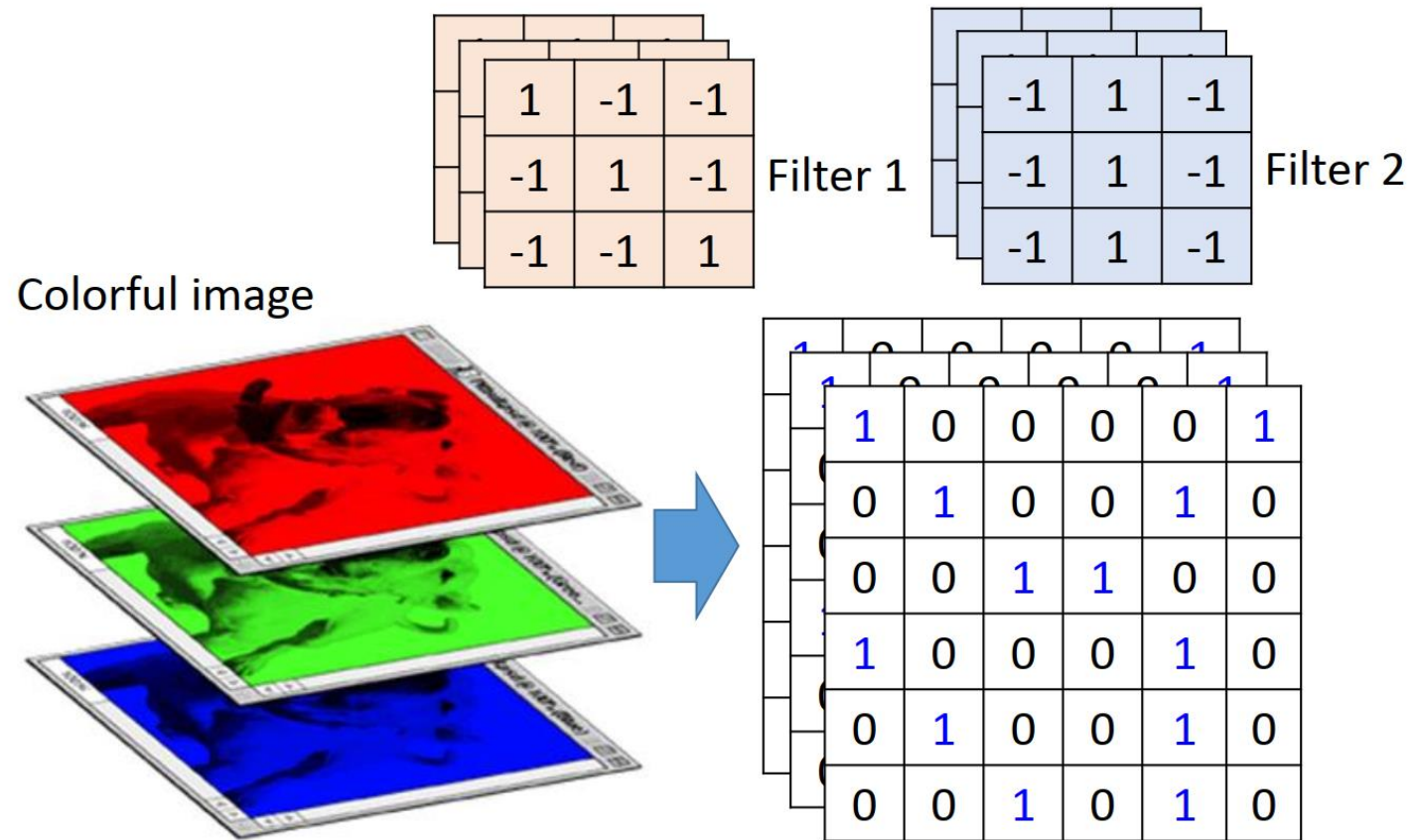
- 6x6 image converted into 4x4 image
- Less parameters for CNN



Feature Map for Color Image



CNN – Colorful image

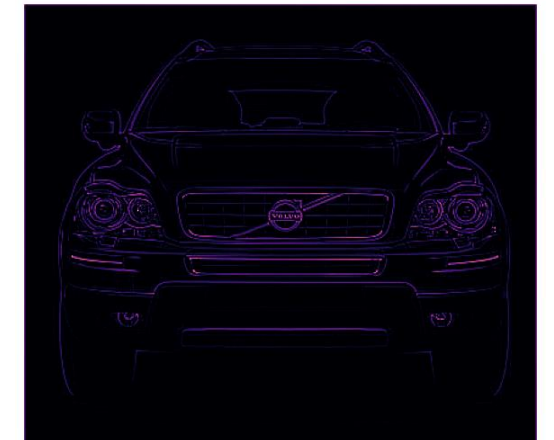
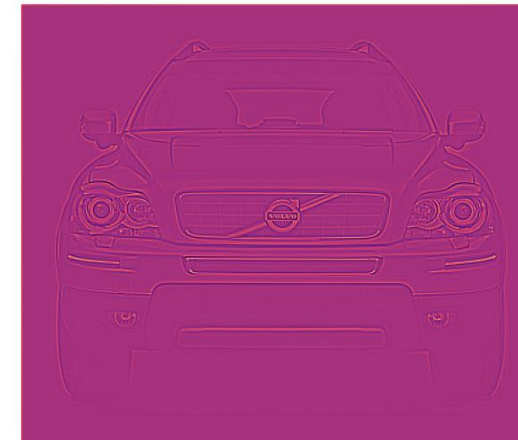
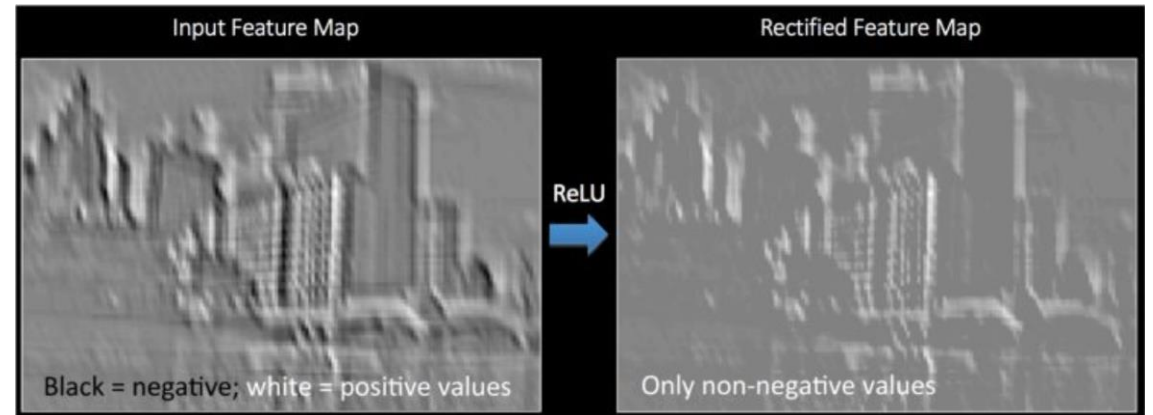


Activation Functions



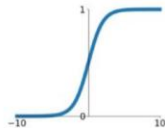
- **ReLU (Rectified Linear Unit):**
 - This is the most commonly used activation function in CNNs.
 - It returns 0 if it receives any negative input, but for any positive value x , it returns that value back.
- ReLU introduces the property of nonlinearity to a deep learning model and solves the **vanishing gradients issue**

- **Convolution + ReLU**



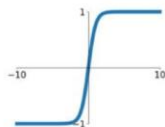
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



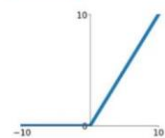
tanh

$$\tanh(x)$$



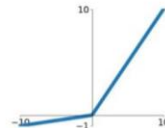
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

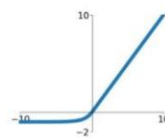


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

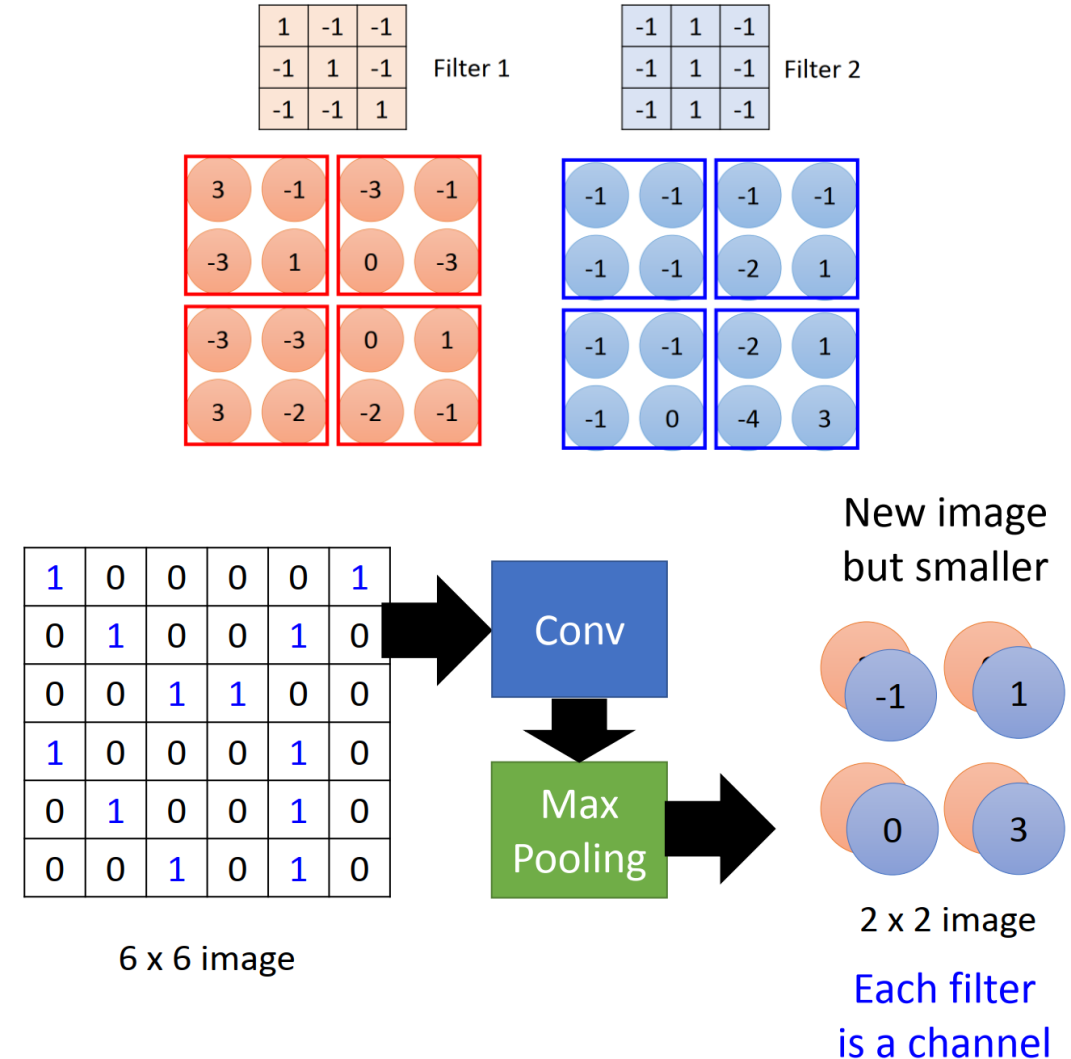


- Pooling layers are used to reduce the dimensions of the feature maps. Thus, it reduces the number of parameters to learn and the amount of computation performed in the network.
- It controls overfitting and **extract dominant features**.
- The pooling layer summarizes the features present in a region of the feature map generated by a convolution layer.
- The most common types of pooling operations are **max pooling** and **average pooling**.

Max Pooling - Example



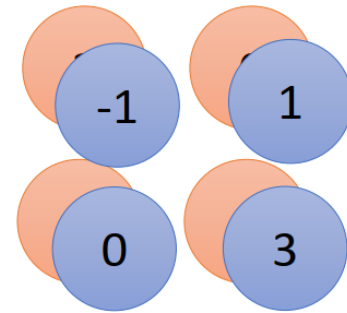
- The pooling layer works by dividing the input feature map into a set of non-overlapping regions, called pooling regions.
- Each pooling region is then transformed into a single output value, which represents the presence of a particular feature in that region.



Convolution and Pooling



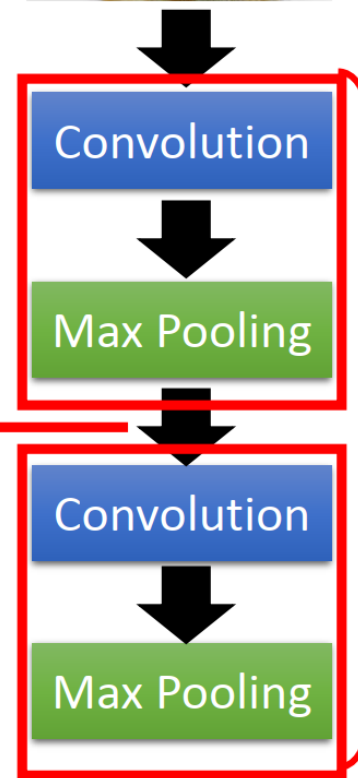
The whole CNN



A new image

Smaller than the original image

The number of the channel is the number of filters

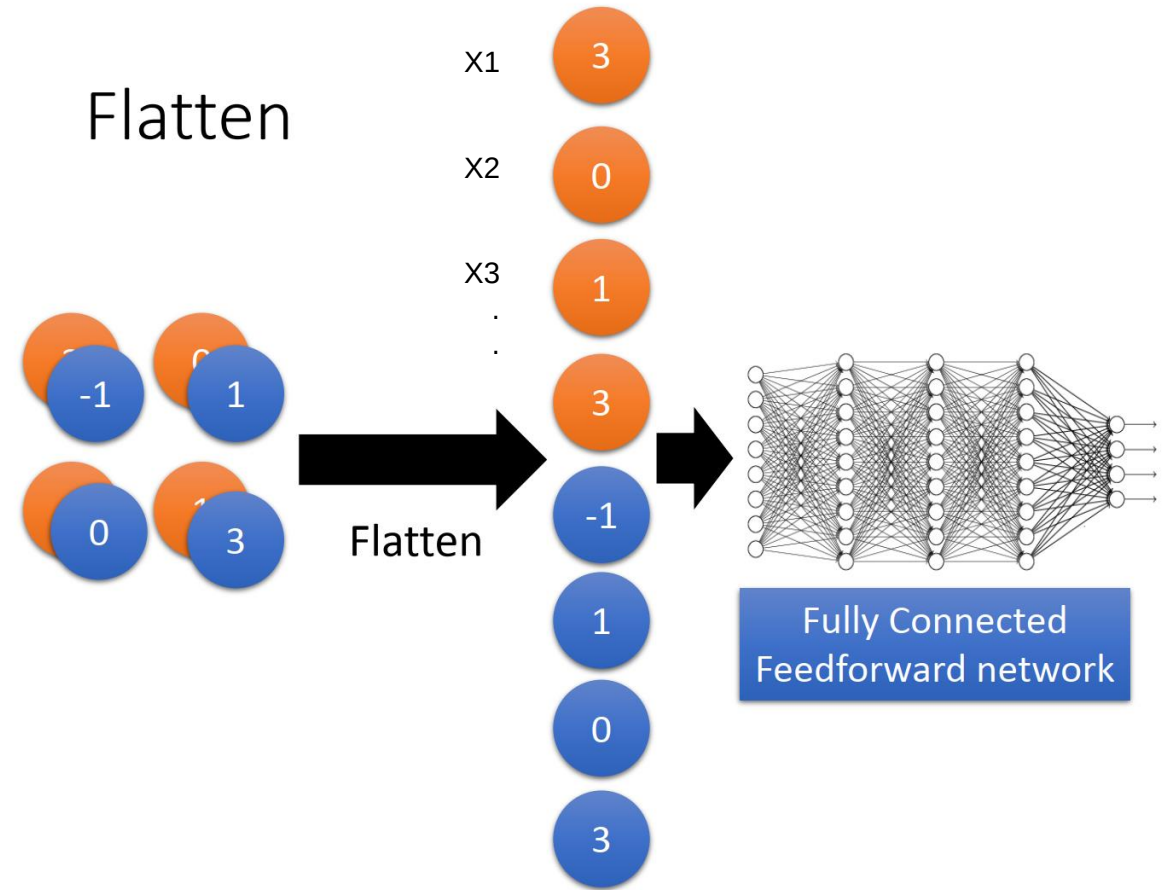


Can repeat many times

Flatten Layer



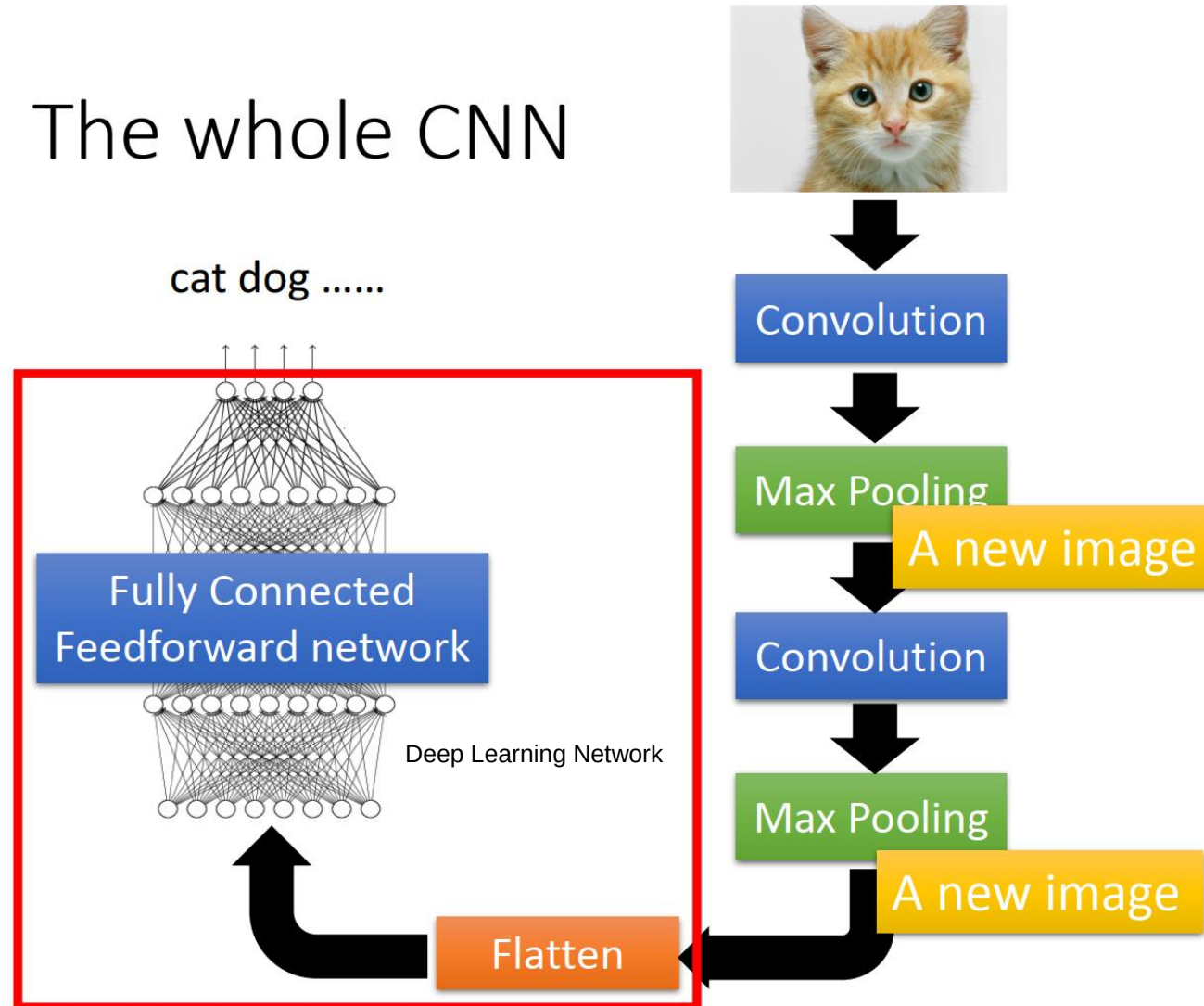
- Flattening is used to convert all the resultant 2-Dimensional arrays from pooled feature maps into a single long continuous linear vector.
- The flattened matrix is then fed into the fully connected layer to classify the image (or any other use-case)



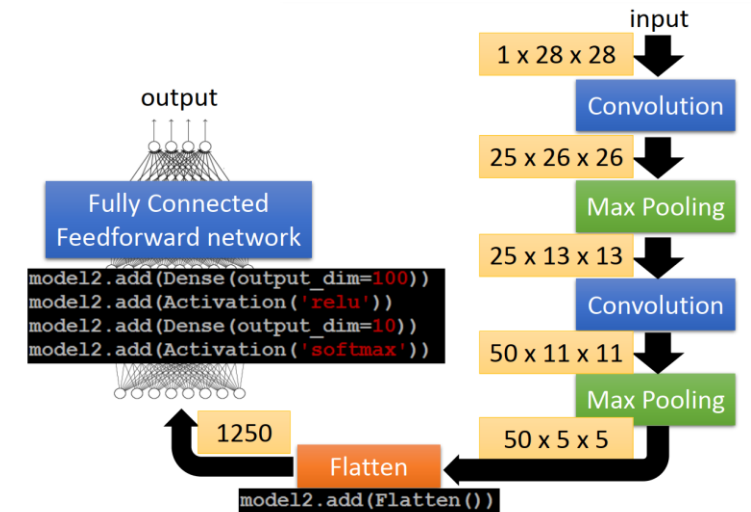
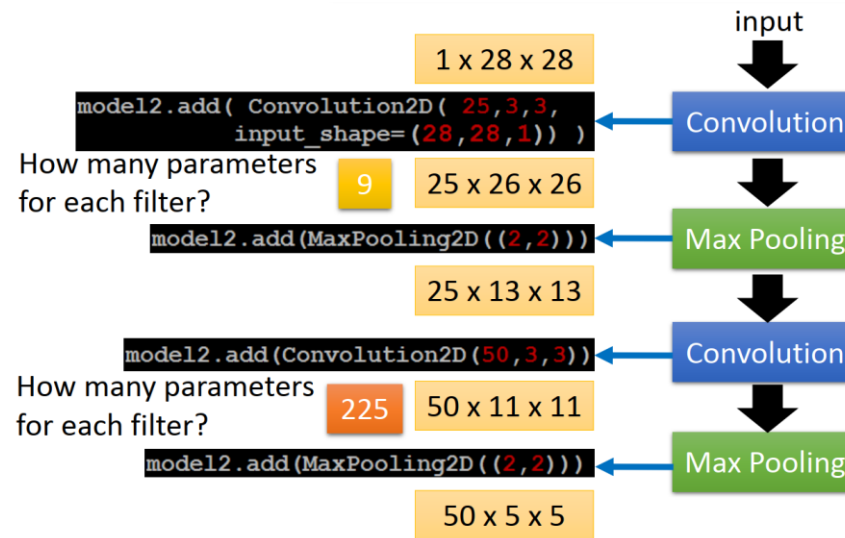
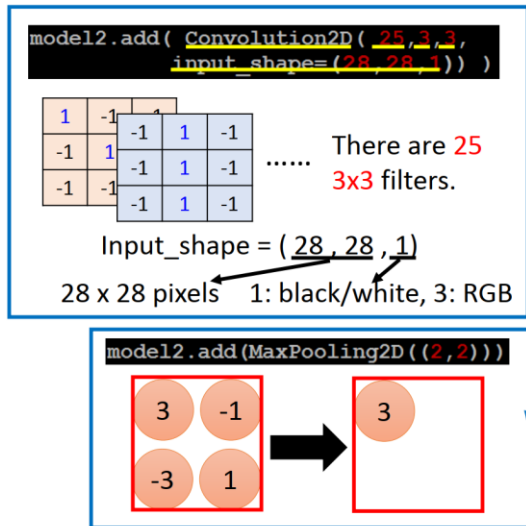
The Final CNN



The whole CNN



- Keras Python for Image Processing



End of Part 4

